

User Mode Linux Redesign

Architecture, Implementation State, and Validation Evidence

Reporting cutoff: branch `umlctl-deploy`, commit `485eff5c96be`

May 14, 2026

Abstract

This report summarizes the User Mode Linux redesign branch as of May 14, 2026. It is a state report for maintainers: the first pages state the branch claim, the delta from inherited UML, and the current validation position; later sections provide design detail, benchmark data, validation evidence, and remaining work. The central claim is that the redesign has moved beyond paper architecture: backend abstraction, static-key gates, profile work, the Rust launcher, and a substantial `kvm-v2` backend are present in-tree. The remaining formal gate is sustained validation and upstream packaging.

1 Executive Summary

1.1 Goal and Current Position

This branch proposes a profile-driven direction for UML while preserving the existing contract: the upstream Linux kernel built as `arch/um` and executed as a host process. The target is not another virtual machine monitor, not a container runtime, and not a Linux reimplementation. The branch-local work is the combination of swappable execution backends, runtime instrumentation gates, profile-driven builds, and validation artifacts from one source tree.

Branch claim. One kernel tree can produce fast production UML, sanitizer-heavy research UML, fuzzing UML, sandbox UML, library-mode UML, and time-travel UML.

Architecture. Layer 1 abstracts the trap mechanism with `struct um_backend_ops`; Layer 2 places static-key gates on hot paths; Layer 3 composes compile-time sanitizers and profile features.

Current state at May 14, 2026. Backend abstraction, static-key infrastructure, profile work, `umlctl`, selftests, and `kvm-v2` are in-tree. `kvm-v2` has working guest entry, syscall dispatch, exceptions, signals, SMP state repair, and post-gadget performance; sustained validation is in progress.

Evidence. `kvm-v2` matches `seccomp` on the substrate gate, reaches cpython-parity 21/21 under UP and SMP, passes 200/200 `kvm-v2` iterations in the first post-T57 two-hour soak, and is faster than `seccomp` on all eight benchmark hosts.

Figure 1: The report in four claims. Each claim maps back to a tracked source document or measured artefact.

1.2 Existing UML Contract and Branch Questions

The branch assumes the existing UML contract: Linux itself runs as a userspace process, remains debuggable and rootless, and can host isolated Linux instances without emulating a full machine [1, 2, 3]. The proposed changes keep that contract and concentrate on branch-local pressure points: userspace trap cost, multiprocessor state ownership, profile composition, and validation breadth.

1.3 Cutoff Snapshot

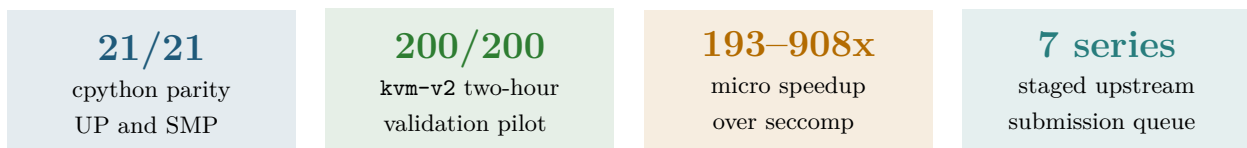


Figure 2: Dashboard metrics at the reporting cutoff. The numbers are intentionally sparse: each is backed by a source table later in the report.

Table 1: Existing UML contract mapped to the branch-local response.

Existing property	Constraint carried forward	Branch-local response
Kernel as a normal host process	Preserve rootless development, CI usability, and reproducible debugging where nested virtualization is unavailable.	Keep stock-host execution through ptrace/seccomp, but add KVM as a fast backend when available.
Isolated Linux instances	Preserve cheap Linux environments without new hardware or privileged host setup.	Make sandbox a profile, not the whole architecture; choose minimal code, seccomp-only defaults, and explicit validation.
Faithful upstream kernel	Preserve Linux behavior rather than introducing a userspace reimplementaion with semantic drift.	Build the upstream kernel as UML; profiles change defaults and instrumentation, not kernel semantics.
Debuggable experimentation	Preserve normal UML debugging value while adding tracing, coverage, and replay hooks as explicit features.	Layer 2 static keys make observability a runtime decision; Layer 3 composes sanitizers by profile.
Known hard problems	Treat trap cost and SMP state ownership as implementation and validation problems, not as presentation claims.	<code>kvm-v2</code> and sustained validation directly target syscall cost, SMP state ownership, parity, and long-run reliability.

1.4 What Is Done, What Is Not

1.5 First Three-Page Takeaway

The branch is best understood as a transition from one default execution story to an explicit profile matrix. The common lower layer is a backend contract; production can choose KVM, CI can choose `seccomp`, legacy hosts can choose ptrace, and each profile can decide whether it wants no hooks, runtime hooks, or heavy compile-time instrumentation.

The current branch has crossed the highest-risk implementation path: `kvm-v2` can run real guest userspace, handle SMP, preserve cross-task state, and beat `seccomp` on measured workloads. The remaining gate is sustained validation: reliability, breadth, and upstreamable series boundaries.

Table 2: Questions the report uses to bound the branch claims.

#	Question	Where answered
1	What branch-local value proposition is being proposed?	Branch claim, non-goals, existing-contract table, and profile matrix.
2	What is the boundary relative to QEMU, Firecracker, gVisor, LKL, and containers?	Non-goals, design boundary, and adjacent-work comparisons.
3	What single architecture lets fast production, research, fuzzing, sandbox, library, and time-travel coexist?	Three-layer architecture and profile sections.
4	What has actually landed, and what is still only planned?	Current-state tables, source map, and upstream queue.
5	Does <code>kvm-v2</code> preserve Linux-visible state across SMP, signals, FPU/XSAVE, faults, and task switches?	KVM design, state-audit method, recent-fix table, and validation ladder.
6	Does the performance win survive beyond a microbenchmark?	Micro, Python, and stress-tier benchmark sections.
7	What validation bar remains before upstreaming?	Sustained validation, Tier 2/3/LTP, and next-work sections.
8	What should maintainer review put pressure on next?	Open-state list, remaining risks, and upstream-path sections.

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Table 3: Reporting snapshot.

Reporting date	May 14, 2026
Cutoff commit	485eff5c96be on umlctl-deploy
Project corpus	253 files under Documentation/virt/uml/redesign/; 32 top-level tools/testing/selftests/um/ test areas; arch/um/backend/kvm-v2/ implementation present
Functional headline	kvm-v2 has working guest entry, syscall dispatch, exceptions, signals, SMP state repair, and post-gadget performance; formal 24-hour validation remains open
Parity headline	cpython-parity is 21/21 under both UP and SMP; substrate gate is PASS=25/FAIL=3/XFAIL=3, matching seccomp
Performance headline	post-gadget kvm-v2 beats seccomp on all eight lab hosts: micro 193-908x, Python 2.99-4.06x, stress 3.71-8.72x
Validation headline	first post-T57 two-hour soak: kvm-v2 200/200 PASS; aggregate 397/400 PASS, with the three failures isolated to iocheck/seccomp timeout behavior

Table 4: Status by major workstream.

ID	Workstream	State	Summary
A	Backend abstraction	DONE	um_backend_ops contract, ptrace/seccomp split, KUnit contract, and documentation landed.
B	Static-key hot paths	DONE	Hot-path gates, debugfs controls, and runtime flip demonstration landed.
C	Profiles and gaps	MOSTLY	Defconfigs and major sanitizer/tracing ports are largely landed; snapshot, syzkaller, and some resumed features remain.
D	KVM backend	IN PROGRESS	kvm-v2 is functional and performant; 24-hour validation, Tier 2/3, LTP, and upstream packaging are the active edge.

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2 Goal and Vision

2.1 Problem Statement

The branch starts from the existing UML contract: Linux itself is compiled to run as a userspace program, with real kernel semantics and a deployable target in environments where full hardware virtualization is not available. The branch-local problem is to add a reviewable structure for performance, feature composition, and operational validation without changing what UML means.

The proposed bar for this branch is:

- production UML should approach native syscall cost when the host exposes KVM;
- research UML should ship with sanitizers, tracing, kprobes, ftrace, BPF, KGDB, and debug surfaces composed from one tree;
- fuzzing UML should provide KCOV, snapshot/forkserver startup, and a path toward a syzkaller vm/uml backend;
- sandbox UML should compile out debug surfaces and minimize host attack surface;
- time-travel and record/replay should become a profile, not a niche configuration footnote.

2.2 Non-Goals

The design boundary is explicit. It does not reimplement Linux in another language. gVisor is useful prior art for backend structure and KVM execution, but UML keeps Linux semantics by building Linux itself [4]. It is not a microVM competitor to Firecracker [5]; the problem here is kernel research and fuzzing fidelity, not cloud cold-start packaging. It is not a unikernel-only library effort either, although the library profile borrows from LKL where that model is useful [6].

2.3 Success Criteria

The branch succeeds when a developer can select a profile, build it from the same tree, and get an artifact whose performance and instrumentation properties match its intended use:

Table 5: Target profile outcomes.

Profile	Intended outcome
prod-fast	KVM backend, no hooks, near-native syscall path.
prod-with-hooks	Fast production binary with hooks compiled but off; tracing can be enabled after a crash without rebuilding.
research	Debug-friendly UML with tracing, kprobes, sanitizers, and interactive control surfaces.
fuzz / fuzz-deep	KCOV, snapshots, sanitizer coverage, replay-oriented hooks, and syzkaller integration.
sandbox	Minimal debug surface and minimized host-side trust boundary.
library	LKL-style direct-call mode for harnesses that can sacrifice ring-split fidelity.
time-travel	Deterministic time and replay hooks as a first-class shipped configuration.

2.4 Why Now

The timing is tied to concrete prerequisites: seccomp mode exists, SMP support exists, and kernel instrumentation is mature enough to compose around static keys and profile-specific Kconfig. Adjacent projects also supply useful models: gVisor for backend/platform design, crosvm for host process

decomposition [7], syzkaller for fuzzing expectations [8], and upstream kernel policy for AI-assisted development disclosure [9].

3 Current State

3.1 Source Basis

The report is grounded in the branch’s source material: goal and plan documents, landed code, the bug diary, current status, and measured evidence. The main source classes are:

Table 6: How the project corpus maps into this report.

Source class	Representative paths	Use in the report
Plan and vision	00-vision.md, README.md, report-presentation/PLAN.md	Defines the goal, non-goals, architecture narrative, and publishing shape.
Architecture	01-architecture/, 02-workstreams/	Defines layers, workstreams, dependencies, profiles, and sequencing.
Code	arch/um/include/shared/backe nd.h, arch/um/backend/kvm-v2/, tools/uml/uml-launcher/	Confirms that the report is describing landed implementation, not only design.
Diary and state	STATUS.md, state-audit/, sustained-validation memos	Provides current truth, bug chronology, closure criteria, and remaining work.
Evidence	tools/testing/selftests/um/, benchmark and soak memos	Supplies measured performance, parity, soak, and gate data.

3.2 Inherited UML Versus This Branch

The attribution boundary matters. I checked it by diffing `origin/master...HEAD` across `arch/um`, `tools/uml`, `tools/testing/selftests/um`, and `Documentation/virt/uml`. That diff spans 688 UML-related files, with 154,501 insertions and 823 deletions. The number is a scope check, not an authorship claim: the existing UML model, the Linux kernel, and generic KVM facilities remain inherited foundations.

Table 7: Attribution boundary for the current branch.

Category	What the report means
Inherited UML and Linux	Linux-as-a-host-process, the upstream UML port, legacy ptrace execution, existing kernel subsystems, the host KVM API, and the general idea of deterministic time-travel are not claimed as new work here.
Recent upstream prerequisites	Seccomp-mode UML and upstream SMP work are treated as enabling context. The redesign builds on them; it does not present them as branch-local inventions.
Redesigned in this project	The explicit backend-ops contract, static-key hook model, profile matrix, validation ladder, staged upstream queue, and <code>kvm-v2</code> state-ownership method are the project architecture.
Implemented in this branch	<code>arch/um/backend/kvm-v2/</code> , backend contract code, profile configs and docs, snapshot/forkserver v1, the Rust launcher and <code>umlctl</code> , KVM/soak/benchmark selftests, and the report/deck workspace are branch-local artefacts.
Planned or paused	KVM-aware snapshot readiness, full record/replay, time-machine workflow, syzkaller <code>vm/uml</code> , LTP curation, and the large upstream <code>arch/um</code> series remain future work.

3.3 Landed Versus Open

The branch contains substantial code and test infrastructure:

- `arch/um/include/shared/backend.h` defines the backend contract used by host-side and kernel-side translation units.
- `arch/um/backend/seccomp/` and ptrace integration provide the non-KVM backends.
- `arch/um/backend/kvm-v2/` contains the active `kvm-v2` implementation, including vCPU, memory, syscall, exception, region, state trace, and LSTAR gadget code.
- `tools/uml/uml-launcher/` and `umlctl` provide host-side launch, supervision, gate, gate-loop, history, and deployment machinery.
- `tools/testing/selftests/um/` contains benchmark, parity, gate, soak, snapshot, KVM, ftrace, kprobes, KMSAN, and process reproducer coverage.

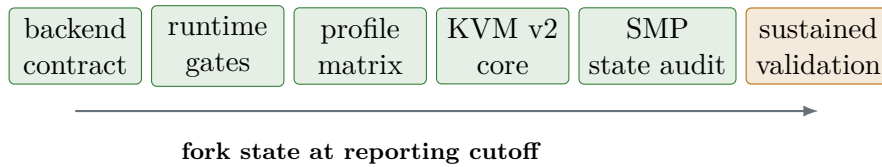


Figure 3: High-level milestone state. Most architectural and implementation work has landed; sustained validation is the active formal gate.

3.4 Current Functional Reading

Table 8: Functional status from `STATUS.md`.

Surface	Reading	Interpretation
<code>init=/bin/echo</code>	UP + SMP yes	Basic guest entry works under <code>kvm-v2</code> .
Python smoke	UP + SMP yes	<code>python3 -V</code> , <code>hashlib</code> , <code>ssl</code> , and single imports work.
Substrate gate	PASS=25, FAIL=3, XFAIL=3	<code>kvm-v2</code> matches <code>seccomp</code> on the curated substrate gate.
<code>cpython-parity</code>	21/21	The old P0, Python C-extension import crashes, is functionally closed.
<code>mt-mini T=8, ncpus=4</code>	30/30 or better in current runs	SMP memory stress reaches the current acceptance bar.
<code>threaded-fork-malloc</code>	0/24000 child failures	Cross-task XMM/FPU guarantee survived the T55/T57 fixes.
<code>wide cpython-parity</code>	134 parity, 0 regression	Larger curated module sweep found no <code>kvm-v2</code> regression.

3.5 Open State

The important open items are bounded:

- Sustained validation must still deliver a formal 24-hour continuous run, Tier 2/Tier 3 workload templates, and LTP curation.
- Snapshot and record/replay work is unblocked but paused behind sustained validation.
- The T57 AVX/XSAVE fix is landed; AVX-512 exposure is deferred to a follow-up if needed.
- The LKML queue is sequenced, but the large `arch/um` series are gated on validation and upstream review strategy.

4 Architecture

4.1 Three Layers

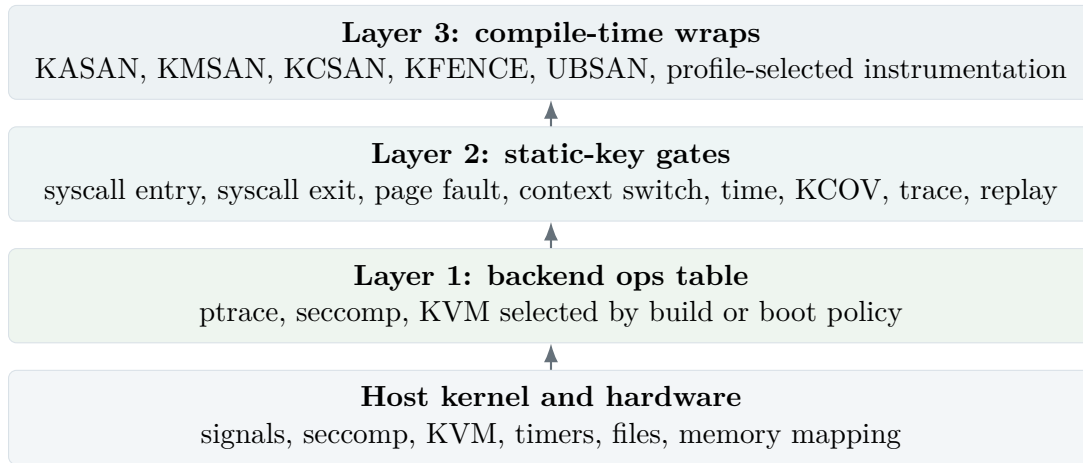


Figure 4: The architectural decomposition. Lower layers do not know about higher-layer policy; profiles choose points in the product space.

Layer 1 isolates the mechanism that makes guest userspace run: `ptrace`, `seccomp`, or `KVM`. Layer 2 isolates optional runtime observation and policy hooks. Layer 3 isolates compiler-level instrumentation whose cost is inherent in the binary. This structure matches the host kernel’s own composition strategy: explicit lower interfaces, static keys for runtime feature toggles, and `Kconfig` for compile-time feature selection.

4.2 Layer 1: Backend Contract

The live contract is `struct um_backend_ops`. It has metadata, capability flags, lifecycle hooks, vCPU execution, memory region callbacks, scheduling hooks, TLB kick support, time hooks, and debug/introspection hooks. The important design choices are:

- every in-tree backend populates the whole table; unsupported cold operations return an explicit error rather than leaving `NULL` fields;
- single-backend builds can compile calls down to direct calls; dynamic builds can select at boot and pay one stable indirect call;
- backend capability flags replaced legacy global booleans, so host-side code asks the selected backend what stub, reaper, and syscall behavior it needs;
- per-mm memory state is owned by the backend, which lets `seccomp` keep its stub model while `kvm-v2` manages memslots and guest mappings.

Table 9: Backend contract categories.

Category	Representative ops	Why it belongs in Layer 1
Lifecycle/trap	<code>probe</code> , <code>init</code> , <code>vcpu_run</code>	Backend availability and guest execution are mechanism-specific.
Memory	<code>mm_create</code> , region add, remove, protect	<code>ptrace/seccomp</code> stubs and KVM memslots have different state models.
Scheduling	thread create, context switch, IPI	Guest scheduling touches backend-owned vCPU or stub state.
Time	clock read, timer set	Determinism and backend timing sources must be abstracted.
Debug	register read/write	KGDB and inspection need a common contract over different mechanisms.

4.3 Layer 2: Static-Key Hot Paths

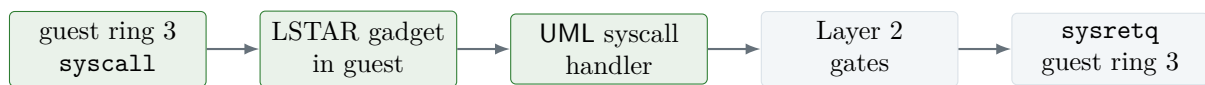
Layer 2 is the answer to a central tension: a research build wants hooks everywhere; a production build wants the hot path cold and small. Static keys make the disabled case a patched NOP sequence and the enabled case an explicit branch to slow-path code. The off-state cost model is therefore stable and auditable.

The report treats Layer 2 as a policy plane rather than a pile of tracepoints. The same hot path can support tracing, KCOV, record/replay, time-travel, and performance event emission, provided each hook has a clear default state and each profile declares which hooks are compiled and enabled.

4.4 Layer 3: Compile-Time Instrumentation

Layer 3 includes KASAN, KMSAN, KCSAN, KFENCE, UBSAN, and related compiler- or allocator-level instrumentation. These features cannot be made free by a runtime branch. They change code generation, memory layout, or allocator behavior. The architecture therefore treats them as profile-level build choices that compose with any backend below.

4.5 A Syscall Through the Stack



fast-path calls stay inside the KVM guest

Figure 5: `kvm-v2` fast syscall shape after gadget revival. Trivial syscalls avoid a host `vmexit` on the hot path.

5 KVM v2 Design

5.1 Model

`kvm-v2` follows the gVisor KVM-platform idea, adapted to UML: guest userspace runs at ring 3 inside a KVM guest, the UML kernel runs at ring 0 inside the same KVM guest, and ordinary syscalls enter the UML kernel through `MSR_LSTAR` instead of trapping to a host monitor [4, 10]. Host exits are still used for operations that truly require host participation, but the target is to keep the common syscall path in guest context.

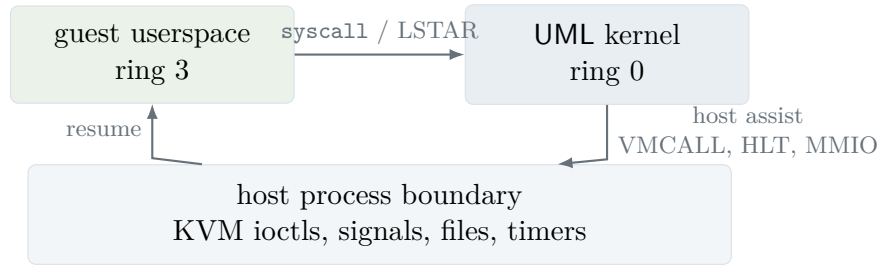


Figure 6: `kvm-v2` execution model. It is not a QEMU-style machine; it is a trap mechanism for UML itself.

5.2 Lessons from the v1 KVM Backend

The `kvm-v2` design is not just a cleaned-up version of the first KVM implementation. The v1 implementation was valuable because it proved the fast trap path was possible, but it also showed that the initial data model would not survive sustained SMP execution. The diary around the April 27 architecture review and the v1 restart guide converged on the same lesson: treating a KVM vCPU as a passive trap resource shared by UML scheduler machinery creates state ownership bugs that cannot be made impossible by adding more repair code.

Table 10: Design lessons carried from KVM v1 into `kvm-v2`.

v1 pressure	Failure mode learned	<code>kvm-v2</code> response
KVM as a faster trap loop	The per-vCPU <code>kvm_run</code> mmap and architectural state have active ownership rules; they are not just a scratch buffer behind a syscall trap.	Make ownership explicit: per-vCPU state, per-task saved state, per-mm state, and synchronization rules are documented and audited.
Repair on next entry	Snapshotting exit state and repairing later does not compose with <code>-EINTR</code> , host signals, and SMP scheduling.	Use explicit entry/exit boundaries, KVM-visible signal masking, and state trace instead of relying on opportunistic repair.
Shadow page-table machinery	Multiple producers had to keep a shadow view coherent while KVM consumed it; every new fix added another ordering rule.	Keep the v2 memory contract smaller and make TLB generation, memslot policy, and per-vCPU refresh rules visible.
Implicit scheduler assumptions	<code>preempt_disable()</code> did not mean “this task cannot move to another host CPU” for the KVM state being touched.	Use <code>migrate_disable()</code> at the dispatch and capture sites that depend on a selected per-host-CPU vCPU.
Ad hoc bug hunting	Individual hypotheses produced regressions or negative results because the state space was not enumerated.	The state-audit method enumerates state items, operations, owners, save/restore rules, and suspect registers before accepting a fix.

This is the changed contract for maintainer review: v2’s core claim is not only that KVM can run UML faster, but that each architectural state item has a declared home and a declared transition rule.

5.3 Implementation Themes

The current `kvm-v2` design is organized around five hard problems:

1. **Guest entry and return.** The backend must enter guest ring 3, handle `SYSCALL/SYSRET` transitions, and recover when host signals interrupt critical regions.

2. **Per-vCPU and per-task state.** General registers, CR2, segment bases, FPU/XSAVE, pending events, IST frames, and KVM run state cannot leak across tasks or host CPUs.
3. **Memory and TLB state.** UML page-table changes must be reflected into KVM-visible mappings and cross-vCPU TLB state without stale translations.
4. **SMP correctness.** Host scheduling, guest scheduling, signal delivery, and KVM vCPU ownership must compose without assuming that `preempt_disable()` pins a host thread.
5. **Performance.** The backend must preserve the win from in-guest dispatch while avoiding correctness regressions from lazy state handling.

5.4 Design Detail: Main Control Surfaces

Table 11: `kvm-v2` implementation surfaces to understand before reviewing patches.

Surface	Representative code	Design role
Initialization and capabilities	<code>init.c</code> , <code>ops.c</code> , <code>Kconfig</code>	Probe host KVM support, wire backend ops, expose configuration choices.
vCPU execution	<code>vcpu.c</code>	Owns <code>KVM_RUN</code> , register marshaling, migration pinning, FPU capture, dispatch, and per-vCPU state.
Syscall path	<code>syscall_trap.c</code> , <code>lstar_gadget.S</code>	Handles LSTAR entry, in-guest fast syscalls, slow-path exits, and return-state repair.
Memory and regions	<code>memslot.c</code> , <code>region.c</code> , backend memory ops	Maps UML memory changes into KVM-visible regions and memslot policy.
Exceptions	<code>exception.c</code>	Installs and handles IDT/TSS/IST paths, page faults, and special recovery paths.
State tracing	<code>state_trace.[ch]</code>	Gives low-overhead, static-key-controlled visibility into state ownership bugs.
Tests	<code>test_marshall.c</code> , <code>test_byteshape.c</code>	Protect register ABI, gadget byte shape, and internal assumptions.

5.5 Design Detail: Fast Path Versus Slow Path

The backend now has two different syscall cost classes. Trivial syscalls covered by the LSTAR gadget stay inside the KVM guest and return directly to guest userspace. Syscalls outside the gadget set, or paths requiring host assistance, still exit to the host and run the full backend machinery. The project should keep these paths described separately:

- **Fast path:** `getpid`-family and clock-like calls that can be answered from guest-visible state with preserved ABI registers and no host-side work.
- **Slow path:** `mmap`, `munmap`, `brk`, I/O, scheduling, signal, and fault paths that need backend state or host services.
- **Recovery path:** interrupted LSTAR, page-fault stub, `#NM`, and signal delivery edges where the backend must rebuild user-visible architectural state exactly.

5.6 State-Audit Method

The state-audit directory is part of the design, not just project diary. It converted a vague SMP corruption class into a finite audit: 149 state items, 50+ operations, an ownership matrix, suspect register audits, runtime tracing, and ranked findings.

Table 12: State-audit layers.

Layer	Artefact	Contribution
1	State inventory	Enumerates register, control, segment, MSR, FPU, event, descriptor, per-task, per-vCPU, and per-mm state.
2	Operations inventory	Lists hardware, KVM, backend, scheduler, and signal operations that read or write state.
3	Ownership matrix	States who owns each state item at each transition.
4	Suspect register audits	Tracks RBP, RAX, RDX, FS_BASE and similar corruption signatures.
5	Toolkit	cscope, ripgrep, ast-grep, bpftrace, ftrace, and parser recipes.
6–8	Findings and fixes	Converts candidates into patches and regression matrices.

The most important fix from this arc was the SMP-T13 `migrate_disable()` finding. UML was using `preempt_disable()` to pin access to per-host-CPU vCPU state, but the build did not have preemption count semantics, so the call did not actually prevent migration. Replacing that assumption with `migrate_disable()` removed a structural cross-vCPU ownership violation.

5.7 Recent Fixes

Table 13: Selected recent `kvm-v2` correctness and performance closures.

Item	Closure	Why it matters
SMP-T41	Recover user RAX in EINTR-mid-PF-stub	Closed the true mt-mini byte-zero residual.
SMP-T54	SOCK_CLOEXEC and <code>umlctl sysrq halt</code>	Removed worker fd leak and init shutdown hang class.
SMP-T55	Per-vCPU FPU-dirty epoch	Restored perf-py-startup gate while preserving cross-task FPU guarantees.
SMP-T57 AVX/XSAVE	CR4.OSXSAVE, XCR0=0x7, CPUID AVX/XSAVE unmask	Turned stress-ng AVX <code>#UD</code> from a fragile mask problem into supported guest architectural state.
Gadget revival	LSTAR in-guest fast path	Collapsed trivial syscall cost from host-exit scale to roughly 90 cycles on measured hosts.

5.8 Design Boundary

The current backend should be presented upstream as a backend for UML, not as a general VMM. The VMM-like machinery exists only to make UML execution fast and faithful. That framing matters: it keeps device emulation, boot firmware, hardware discovery, and cloud microVM expectations out of scope.

Table 14: Shipped profile intent.

Profile	Backend default	Hooks	Sanitizers	Purpose
<code>prod-fast</code>	KVM > seccomp	none	none	Minimum overhead production target.
<code>prod-with-hooks</code>	KVM > seccomp	compiled, off	none	Production artifact with post-failure observability.
<code>research</code>	seccomp	trace, kprobes	KASAN, UBSAN	Debug and exploit-reproduction profile.
<code>fuzz</code>	seccomp	KCOV, snapshot	KASAN	syzkaller and harness fuzzing.
<code>fuzz-deep</code>	seccomp	KCOV, snapshot, replay	KASAN, KCSAN	deeper race/replay investigations.
<code>sandbox</code>	seccomp-only	none compiled	none	Minimized TCB and debug surface.
<code>library</code>	direct call	n/a	optional KASAN	LKL-style harness integration.
<code>embedded</code>	ptrace	none	none	Old-host fallback.
<code>time-travel</code>	seccomp	trace, time-travel	KASAN	deterministic clock and replay.

6 Profiles and Validation

6.1 Profile Matrix

6.2 Validation Ladder

6.3 Sustained Validation Pilot

The three failures were isolated to `iocheck/seccomp` timeout behavior under two-worker contention. They were not data miscompares and not a `kvm-v2` regression. The meaningful reading is that daemon mode, threshold-trip behavior, Wilson summary generation, and sustained `kvm-v2` workload rotation all worked.

7 Performance and Benchmarks

7.1 Benchmark Method

The post-gadget cross-host benchmark used one kernel build, one benchmark bundle, the same orchestrator, performance governor, and the same three tiers across eight lab hosts:

- **micro**: `getpid-loop`, `RDTSC-bracketed`, pure trap-mechanism cost.
- **py**: canned Python startup/mixed-syscall workload.
- **stress**: `mt-mini SMP T=8, ncpus=4`, strict memory verification.

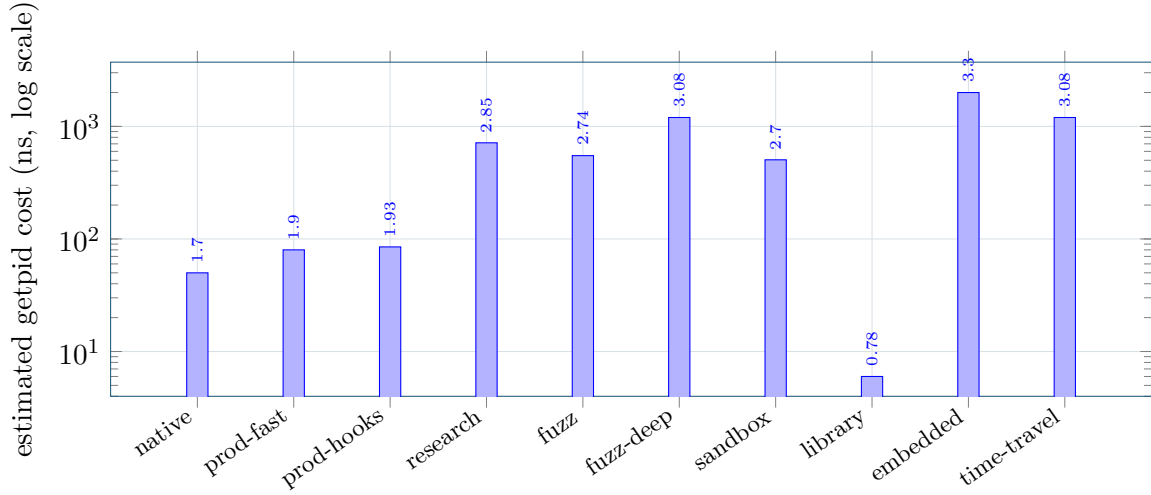


Figure 7: Profile cost model from the redesign docs, anchored to a 50 ns normal Linux `getpid` baseline. The value of profiles is the ability to span direct-call library mode through heavily instrumented research and time-travel builds without forking the tree.

tooling builds: launcher and <code>umlctl</code>
kernel builds: selected profile compiles and links
internal contracts: KUnit, marshal, byte-shape checks
substrate parity: PASS=25/FAIL=3/XFAIL=3, matching <code>seccomp</code>
process/syscall classes: fork, exec, mmap, signals
performance gates: prevent fast-path regression
real workload boot: Python and stress workloads
sustained validation: 24-hour run plus Tier 2/3 and LTP curation

Figure 8: Validation ladder. Short gates protect the inner loop; sustained validation is reserved for milestone candidates.

7.2 Native Baseline Context

The cross-host charts below use same-host speedup over `seccomp` because that ratio cancels most CPU-frequency and microarchitecture noise. For intuition, the profile cost model also carries a normal Linux `getpid` reference of 50 ns, shown here as 1.0x.

7.3 Micro Tier

The micro result is the clearest evidence that the LSTAR gadget changed the backend’s cost class. Before the gadget, a trivial syscall paid a host-exit round trip. After gadget revival, the measured `kvm-v2` cost is roughly 87–153 cycles on the tested machines, while `seccomp` remains tens of thousands of cycles per call.

7.4 Application and Stress Tiers

The Python tier matters because it turns the micro win into an application-level win: 2.99–4.06x faster than `seccomp`. The stress tier matters because it combines performance with correctness: the run reported zero strict memory failures and zero verify failures on every host.

Table 15: First post-T57 sustained-validation pilot, 2026-05-14.

Workload	Backend	n	pass	fail	pass rate
memcheck	kvm-v2	60	60	0	100.00%
memcheck	seccomp	60	60	0	100.00%
iocheck	kvm-v2	60	60	0	100.00%
iocheck	seccomp	60	57	3	95.00%
stress-ng	kvm-v2	40	40	0	100.00%
stress-ng	seccomp	40	40	0	100.00%
tier1-pylibs	kvm-v2	40	40	0	100.00%
tier1-pylibs	seccomp	40	40	0	100.00%
kvm-v2 aggregate		200	200	0	100.00%
Overall aggregate		400	397	3	99.25%

Table 16: `getpid` context relative to a normal host syscall baseline.

Path	Model cost	Relative to normal	Reading
Normal Linux host syscall	50 ns	1.0x	Human-scale baseline for a trivial syscall on the host.
kvm-v2 prod-fast	80 ns	1.6x	Same cost class as the host baseline for gadget-covered calls.
kvm-v2 cross-host gadget	87–153 cycles	mechanism tier	Raw cycles vary by host; the robust claim is same-host speedup over seccomp .
seccomp profiles	505–1,200 ns	10–24x	Expected for research, fuzz, sandbox, and time-travel profiles where instrumentation or isolation dominates.
ptrace fallback	2,000 ns	40x	Old-host fallback, useful for compatibility rather than speed.
direct library call	6 ns	0.12x	LKL-style mode removes the syscall boundary; it is not a faithful ring-split kernel target.

7.5 Detailed Cross-Host Table

7.6 Performance Interpretation

The performance story has three layers:

- **Mechanism win.** The micro tier proves that in-guest fast-path dispatch removes most seccomp signal/stub cost for trivial syscalls.
- **Workload win.** The Python tier proves that real userland startup benefits from those trivial syscalls becoming cheap.
- **Concurrency win.** The stress tier is mmap/page-fault heavy, so the gadget is less directly relevant; the continuing 3.71–8.72x speedup comes from the broader vCPU and SMP design.

The correct benchmark unit is the ratio against seccomp on the same host, not the raw absolute latency across machines. That choice removes most CPU-frequency and microarchitecture noise and keeps the claim tied to a backend decision.

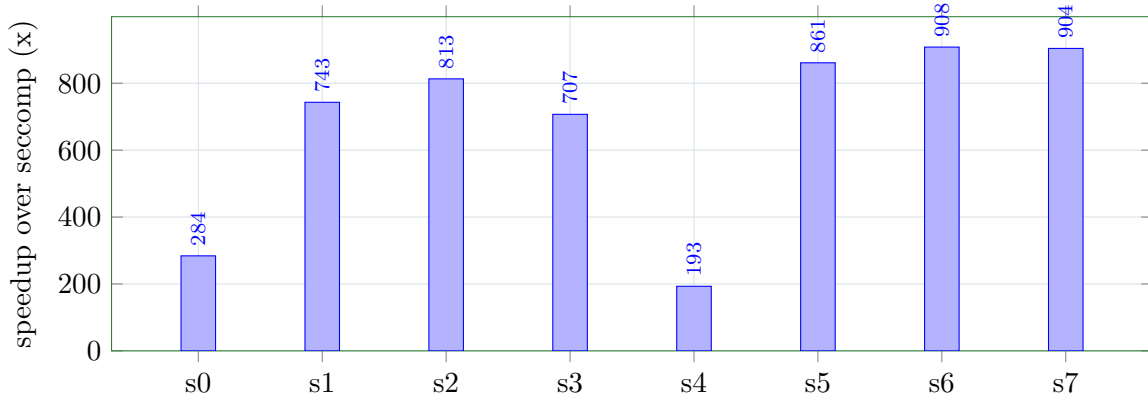


Figure 9: Post-gadget micro speedup. Trivial syscall dispatch is 193–908x faster than seccomp across the eight hosts.

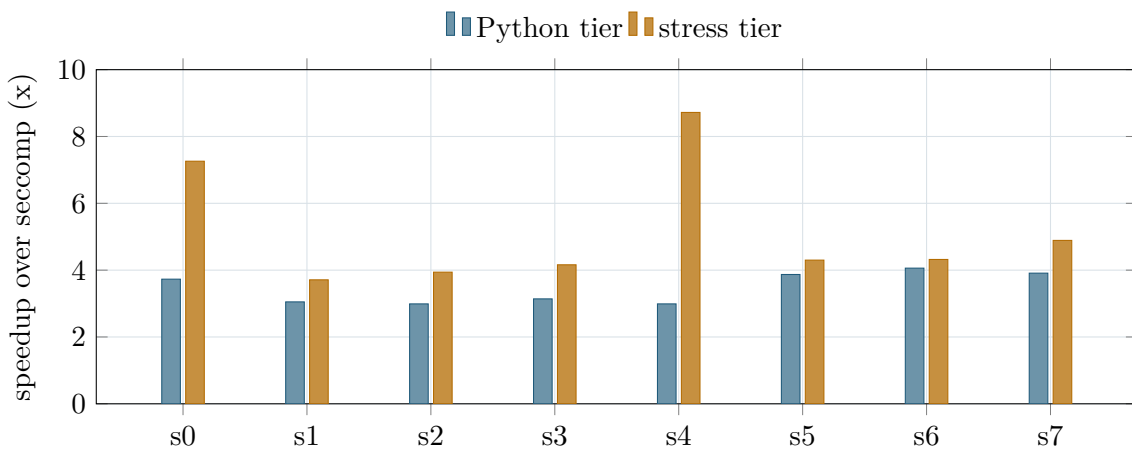


Figure 10: Application and stress tiers. kvm-v2 wins on every host in both tiers.

8 Remaining Work and Upstream Path

8.1 Immediate Work

8.2 Snapshot, Replay, and Time-Travel Status

The “time machine” story should be presented as a staged roadmap, not as a finished product. The branch has real snapshot/forkserver plumbing and an explicit time-travel profile, but full record/replay is still future work.

8.3 Residual Risks

Two narrow residual flake classes are tracked as P2: an mt-mini page-recycling class and a high-yield mt-yieldonly RIP-corruption class. They are not current sustained-validation blockers and do not block snapshot/record-replay resumption, but they should remain visible in the report because they represent exactly the kind of concurrency state drift the state-audit method was built to catch.

8.4 Upstream Queue

The upstream strategy is deliberately staged. Small, independent subsystem patches go first. The backend abstraction is the tentpole. Static-key hot paths and profile work follow. The KVM backend is last because it is the largest technical and review burden.

Table 17: Post-gadget speedup over seccomp.

Host	CPU	micro	Python	stress
s0	i9-12900K	284x	3.73x	7.26x
s1	Xeon E3-1225 v6	743x	3.05x	3.71x
s2	Xeon E3-1225 v5	813x	2.99x	3.94x
s3	Xeon W-2123	707x	3.14x	4.16x
s4	i5-12600K	193x	2.99x	8.72x
s5	Ryzen 7 7840HS	861x	3.87x	4.30x
s6	Ryzen 7 7840HS	908x	4.06x	4.32x
s7	Ryzen 7 7840HS	904x	3.91x	4.89x

Table 18: Next work at the reporting cutoff.

Item	State	Notes
24h sustained validation	IN PROGRESS	Daemon exists and passed a two-hour short run; formal 24-hour run remains.
Tier 2 / Tier 3	IN PROGRESS	Designs exist; templates and bootstrap flow still need implementation.
LTP curation	PLANNED	Needs KEEP/SKIP list and a UML-appropriate runner.
KVM-aware snapshot readiness	PLANNED	Unblocked after Python parity closure; resumes after sustained validation.
Record/replay and time machine	PLANNED	Time-travel profile exists; full replay workflow remains paused.
AVX-512 follow-up	PLANNED	Optional follow-up for the remaining T57 vm-method failures.

8.5 Main Message for Maintainers

The upstream path makes three narrower claims:

1. UML already has multiple backend mechanisms; making the backend contract explicit is a cleanup and an enabling move.
2. static-key hooks are the right kernel-native way to make instrumentation runtime-flippable without charging every profile.
3. the `kvm-v2` evidence is strong enough to justify continued validation and an eventual upstream RFC, but the large KVM series should wait until sustained validation is formalized.

A Source Map

Table 21: Primary source material used for this report.

Path	Role in the report
<code>00-vision.md</code>	Vision, non-goals, success criteria, stakeholder framing.
<code>README.md</code>	Top-level project layout and high-level architecture summary.

Path	Role in the report
STATUS.md	Current state, phase ledger, workstream status, open items.
01-architecture/three-layers.md	Backend/gate/wrap architecture.
01-architecture/data-flow.md	Per-profile syscall cost model and profile comparison.
02-workstreams/README.md	Workstream decomposition and critical path.
02-workstreams/D-kvm-backend/README.md	KVM backend motivation and design framing.
03-architecture-review-2026-04-27/00-synthesis.md	v1 root-cause analysis: KVM state ownership, signal masking, and shadow page-table lessons.
02-workstreams/D-kvm-backend/25-v2-restart-guide.md	Mechanical archive/restart plan that records why v1 was retired and what v2 had to change.
02-workstreams/D-kvm-backend/26-v2-implementation-plan.md	Detailed v2 implementation plan and milestone diary.
02-workstreams/D-kvm-backend/state-audit/	State ownership audit, bug chronology, and recent fixes.
02-workstreams/D-kvm-backend/bench-cross-host-2026-05-04-postgadget.md	Cross-host benchmark data and methodology.
02-workstreams/D-kvm-backend/phase-J-soak-2h-2026-05-14.md	Latest sustained-validation pilot result; filename keeps the historical internal label.
03-profiles/README.md	Profile matrix and estimated per-profile cost model.
05-validation/	Validation tiers, benchmarks, conformance, CI, and quality strategy.
upstream-patches/SUBMISSION-QUEUE.md	Sequenced LKML/upstream queue.
tools/testing/selftests/um/	Concrete gate, parity, soak, benchmark, and reproducer harnesses.
tools/uml/uml-launcher/	Host-side launcher and <code>umlctl</code> machinery.

B Reporting Data Files

The report and deck share normalized data under `Documentation/virt/uml/redesign/report-resentation/data/`. At this cutoff the important files are:

- `bench_speedups.csv` and `bench_ratios.csv` for the eight-host post-gadget benchmark.
- `soak_summary.csv` for the 2026-05-14 two-hour sustained-validation pilot.
- `profile_getpid.csv` for profile cost visualization.
- `workstreams.csv`, `validation_ladder.csv`, `upstream_queue.csv`, and `status_kpis.csv` for status tables.

References

- [1] Jeff Dike. “A User-mode Port of the Linux Kernel”. In: *Proceedings of the 4th Annual Linux Showcase and Conference*. USENIX / ALS 2000. 2000.

Table 19: Snapshot, replay, and time-travel state.

Surface	State	Reading
Snapshot/forkserver v1	DONE	C-09 landed a cooperative <code>um_snapshot_ready()</code> point, AFL-compatible fds 198/199, <code>/sys/kernel/um/state_version</code> , launcher fd plumbing, and <code>snapshot-smoke</code> coverage.
v1 ceiling	known limit	Status is reported as zero, payloads must be short and non-blocking, there is no on-disk snapshot, and v1 is not SMP-capable. These are documented limits, not hidden claims.
KVM-aware readiness	PLANNED	Issue #168 is unblocked by Python parity closure but intentionally sits behind sustained validation.
Time-travel profile	profile exists	<code>uml/time-travel</code> selects <code>seccomp-only</code> , <code>UP</code> , <code>CONFIG_UML_TIME_TRAVEL_SUPPORT</code> , and debug surfaces. The runtime hook gate exists; today its slow path is still a counter stub.
Record/replay/time machine	PLANNED	Issue #169 covers the first-class deterministic replay extension. The event log, replay consumer, and user workflow are not implemented yet.
Syzkaller binding	planned external	The in-tree forkserver surface is pinned; the remaining <code>pkg/vm/uml</code> Go driver belongs in the syzkaller repository.

Table 20: Sequenced upstream queue.

#	Series	Size	Subsystem	Status
1	<code>bpf-hygiene-v1</code>	2 patches	BPF	ready
2	<code>kmsan-arch-callback-rfc</code>	1 RFC patch	mm/kmsan	ready
3	<code>ftrace-notrace-generic-v1</code>	1 patch	tracing/ftrace	prepared
4	backend ops abstraction RFC	about 12 patches	arch/um	to write
5	static-key hot paths	about 6 patches	arch/um	to write
6	per-profile C-series	about 20 patches	arch/um plus sub-systems	blocked by Series 7 ordering
7	KVM backend series	about 15 patches	arch/um plus KVM	blocked by sustained validation

- [2] *User Mode Linux HOWTO*. https://docs.kernel.org/virt/uml/user_mode_linux_howto_v2.html. Linux kernel documentation.
- [3] *User Mode Linux*. <https://user-mode-linux.sourceforge.net/>. Project homepage.
- [4] *gVisor Architecture Guide: Platforms*. https://gvisor.dev/docs/architecture_guide/platforms/. gVisor documentation.
- [5] *Firecracker MicroVM*. <https://firecracker-microvm.github.io/>. Project documentation.
- [6] *Linux Kernel Library*. <https://github.com/lkl/linux>. Project repository.
- [7] *crosvm Book: Architecture Overview*. <https://crosvm.dev/book/architecture/overview.html>. Project documentation.
- [8] *syzkaller*. <https://github.com/google/syzkaller>. Project repository.
- [9] *Linux Kernel Documentation: Coding Assistants*. <https://docs.kernel.org/process/coding-assistants.html>. Linux kernel documentation.
- [10] *KVM API Documentation*. <https://docs.kernel.org/virt/kvm/api.html>. Linux kernel documentation.